

TELANGANA STATE-2016
CLASS 10
GENERAL SCIENCE, Paper – I

Time: 2 Hours 45 min.]

Maximum Marks: 40

Instructions:

1. In the time duration of 2 Hours 45 minutes, 15 minutes of time is allotted to read and understand the Question paper
Answer the Questions under Part-A on separate answer book.
3. Write the answers to the Questions under Part –B on the Question paper itself and attach it to the answer book of Part-A.

Part - A

Instructions:

1. Part - A comprises of three sections I, II and III.
2. All the questions are compulsory.
3. There is no overall choice. However, there is an internal choice to the questions under Section - III.

SECTION-I

NOTE:

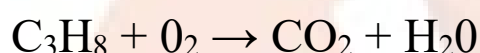
7x1=7

1. Answer all the questions.
2. Each question carries 1 mark.
3. Answer each question in 1 or 2 sentences.

1. Give an example to explain that evaporation is a cooling process.

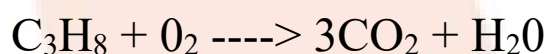
Ans. Sweat or Perspiration is an example to explain that evaporation is a cooling process. Since sweat evaporates heat to cool your body. So the process is actually the amount of heat applied to change liquid into a vapour form.

2. Balance the equation.



Ans. Let us count the number of atoms in the equation above -
Carbon:3 Hydrogen:8 + Oxygen:2 $\rightarrow$ Carbon: 1 Oxygen: 2 +
Hydrogen: 2 Oxygen:1

Lets balance out Carbon. Multiply products side CO_2 by 3



Now, the carbons are balanced, lets focus on hydrogen. Multiply product side H_2O by 4.



The equation gives 8 hydrogen. Next we can check our oxygen. There will be 10 oxygen on the products side and 2 on the reactants so to balance these out, we multiply the O_2 on the reactants side by 5.

The final equation will be $\text{C}_3\text{H}_8 + 5\text{O}_2 \rightarrow 3\text{CO}_2 + 4\text{H}_2\text{O}$.

3. What happens to the image, if a convex lens is made up of two different transparent materials, as shown in figure ?



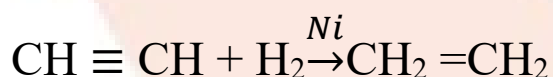
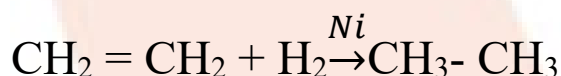
Ans. Given convex lens is made up of two different materials. The two different materials have two different refractive indices. So the lens has two different focal lengths. Hence two different images are formed.

4. $\text{CH}_2 = \text{CH}_2 + \text{H}_2 \xrightarrow{\text{Ni}} \text{CH}_3 - \text{CH}_3$ is an addition reaction.

$\text{CH} \equiv \text{CH} + \text{H}_2 \xrightarrow{\text{Ni}} ?$

Predict and write the products.

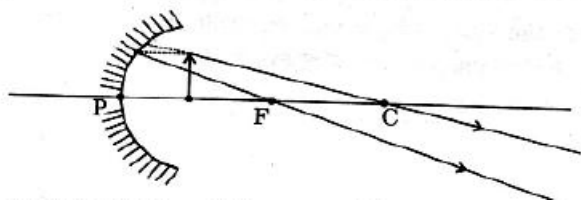
Ans.



This reaction in general is an example of Addition reaction, In which two or more than two reactant reacts together to give single product.

Here in this reaction addition of hydrogen is taking place on the alkene molecule of ethene.

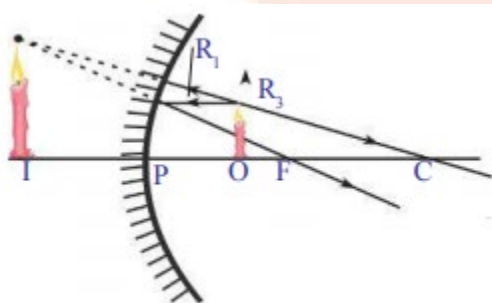
5. Complete the diagram and draw the image..



Ans. We notice that the two reflected rays (Ques Figure) diverge and will not meet. While doing the experiments for a case such as this we were unable to find any place where we get a sharp image on the screen. This ray diagram tells us that since the reflected rays are diverging we will not get an image anywhere. So even if we had moved the screen much away from the mirror, we would not have found an image.

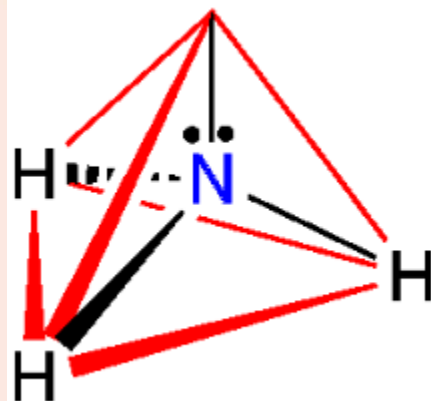
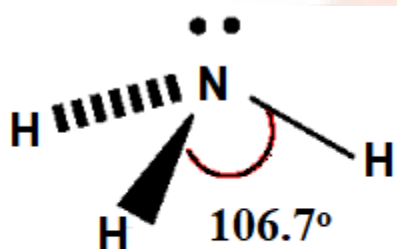
Remember what we did to find the image in a plane mirror. To decide the position of image we extended the reflected rays backwards till they meet. We will do the same here. When we look in the mirror we are looking at these diverging reflected rays. They appear to be coming from one point. We can get this point by extending the rays backwards as shown in figure 14. The image does not really exist the way we see in other cases, but it is visible to us.

As seen in the (Ans figure), the image will be erect and enlarged. This image that we get by extending the rays backwards is called a **virtual image**.



6. Draw the structural diagram of Ammonia molecule as per the valence-shell electron pair repulsion theory .

Ans. Ammonia – 3 Bonds and One Lone Pair. Arranged in tetrahedral shape. HNH angle of 106.7° .



7. Why the soil of agricultural lands get tested for pH ?

Ans. The pH value of the land indicates whether the land is alkaline or acidic, and how acidic or alkaline it is.

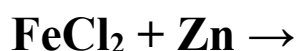
SECTION - II

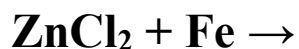
6x2=12

NOTE:

1. Answer all the questions.
2. Each question carries TWO marks.
3. Answer each question in 4-5 sentences.

8. Write the products of given reactions, if any. Give reasons.





This is displacement reaction. As the Zinc is more reactive than the iron therefore it displaces the iron chloride to form its own salt.



As Zinc is more reactive than the iron therefore this doesn't carry any reaction.

Hope this would have been clear to you if not you can contact me on the comment section. Mark as the brainiest.

9. Why Dobereiner, Newlands and Mendeleeff were not 100% successful in their classification of elements? Why the modern table is relatively better classification? Predict the reason.

Ans. Dobereiner, Newlands and Mendeleeff were not 100% successful in their classification of elements because they were classifying on the basis of atomic masses not atomic numbers

Modern periodic table tends to be better as because:-

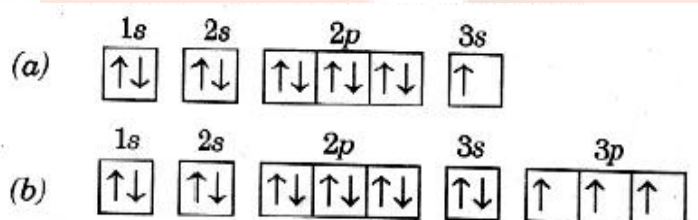
- in modern periodic table isotopes positioned is being justified.
- it has a unique position for hydrogen as it assembles both with halogens and alkali metals
- the elements here are placed according to their increase atomic no.

10. Why water drops (dew) form on flowers and grass during morning hours of winter season?

Ans. Due to the process of precipitation, evaporation and transpiration.

The morning air is cool and cannot hold the moisture in it that's the excess moisture is lost and forms a dew on flowers and grasses.

11. Observe the electronic configurations given below and write the group and period numbers of those elements.



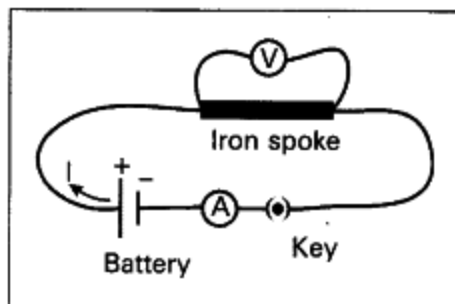
Ans.

(a):- This is the configuration of the **Na** which is in **1st Group** and **3rd period**.

(b):- This is the configuration of the **P** which is in **15th Group** and **3rd period**.

12. Draw the experimental set-up to verify that $\frac{V}{I}$ is constant for a Conductor.

Ans. Aim : To show that the ratio of V/I is a constant for a conductor.



From this we can conclude that potential difference between the ends of the iron spoke is directly proportional to the current passing through it.

This is ohm's law.

$$\frac{V}{I} = \text{Constant}$$

13. Least distance of distinct vision of a person is observed as 35 cm. What lens is useful for him to see his surroundings clearly? Why?

Ans. According to the question the person is suffering from hyperopia, in this problem the power of lens of eyes decrease and we get unable to see the things clearly that are very close to our eyes because the rays that come from the object, pass through the lens of eye meet behind the retina.

These types of problems can be solved by using the convex lens (of appropriate power) because these lens converge the rays. According to the condition of eyes we can find the focus length and appropriate convex lens that is suitable for eyes.

SECTION- III**4x4=16**

NOTE: 1. Answer all the questions..

2. Each question carries 4 marks.

3. There is internal choice for each question.

Only one option from each question is to be attempted.

4. Answer each question in 8-10 sentences.

14. Write the types of Allotropes of Carbon. Give any three examples of each.

Ans. Allotropes of carbon:

The allotropes of carbon are classified into two types.

They are : (I) Amorphous forms

(ii) Crystalline forms

(i) Amorphous forms:- In Amorphous forms the molecules are randomly arranged.

Example: Coal, coke, wood charcoal carbon etc.

(ii) Crystalline forms:- In this forms, molecules are uniformly arranged.

Example: Diamond, graphite and buckminsterfullerene.

OR

Write any 4 characteristic features of homologous series of Organic compounds.

Ans. Characteristics of a homologous series

1. Each member of the series differs from the preceding one by the addition of CH_2 group and by molecular mass of 14 a.m.u.

2. All members of a homologous series share the general formula. (Same elements and same functional group).

For example, the general formula for alkane is $\text{C}_n \text{H}_{2n+2}$, alkene is $\text{C}_n \text{H}_{2n}$,

3. The physical properties of the members change gradually i.e. show graduation in properties as the as the number of carbon atoms per molecule increases.

i.e., as molecular mass increases.

For example melting point, boiling point and the density of the successive members of the homologous series increase with the increase in molecular mass.

Examples:

Alkane Molar mass Boiling point Physical state

(i) Methane 16 -161.4°C gas

(ii) Ethane 30 -88.3°C gas

(iii) Propane 44 -44.5°C gas

(iv) Butane 58 0°C gas

(v) Pentane 72 36.2°C liquid

Members of a homologous series have similar chemical properties. For example: Methane reacts with chlorine to form methyl chloride.



Similarly, ethane reacts to form ethyl chloride



15. An object of height 5 cm is placed at 30 cm distance on the principal axis in front of a concave mirror of focal length 20 cm. Find the image distance and size of the image.

Ans.

$$u = -30 \text{ cm}$$

$$f = -20 \text{ cm}$$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} + \frac{1}{-30} = \frac{1}{-20}$$

$$\frac{1}{v} = \frac{-1}{20} - \frac{(-1)}{30}$$

$$\frac{1}{v} = \frac{-1}{20} + \frac{1}{30} = \frac{-3+2}{60} = \frac{-1}{60}$$

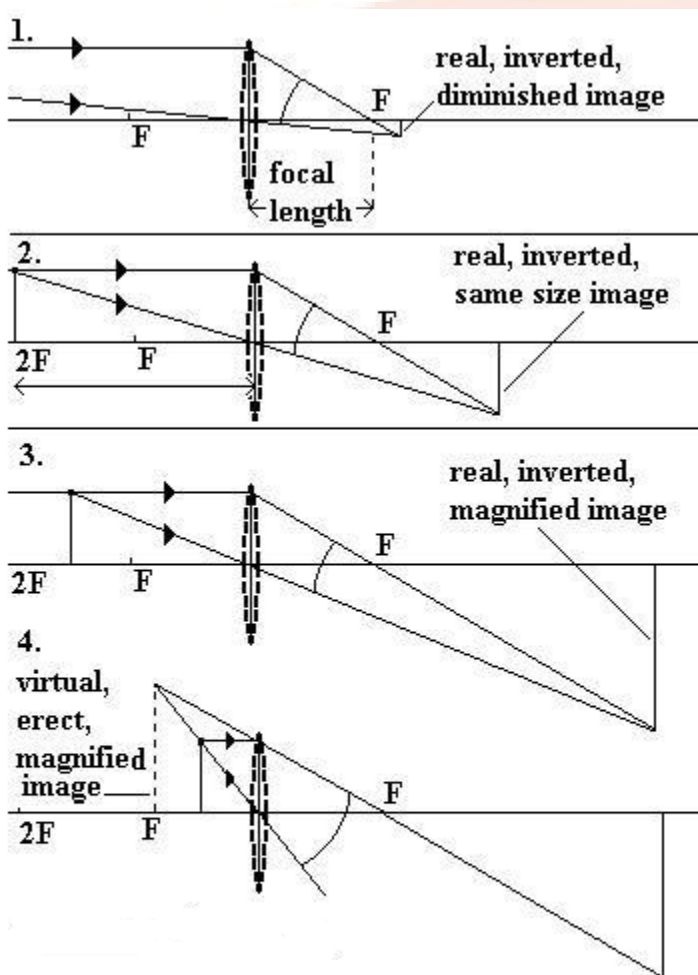
$\therefore$ Image Distance $v = 60$ (-sign)

$$\text{Size of object} = \frac{-60}{-30} = 2 \text{ cm}$$

OR

Explain the behavior of light rays in any four situations of their incidence on a convex lens.

Ans. Ray Diagrams for lenses



16. List out the materials required for the Oersted experiment of Electromagnetism. Write the procedure of the experiment. What do you understand by this experiment?

Ans.

Materials Required

- D battery
- Insulated wire
- Electrical tape
- Compass
- Box
- Electrical tape

Procedure

Take a insulated wire and use electrical tape to secure the wire and then attach the same to one side of D Battery. Make a loop with the wire which can run along the table and ground to reconnect to the battery. As soon as the wire gets connected the current begins to flow. Bring your compass near the flow and record the observations.

The wire carrying the current that will create a magnetic field around itself. When you bring the compass near the wire it will cause the compass needle to move.

OR

What are the materials required for the experiment to show the chemical decomposition of water? Write the procedure of the experiment. Name the products which we get in this reaction.

Ans. Decomposition of water into Hydrogen and Oxygen when an electric current is passed is known as electrolysis of water.

We would be requiring electrolysis apparatus, rubber stopper, one holed, glass tube, 25 cm length, bent at 90° angle in center 1 large test tube

The electrolysis apparatus have 6 volt battery, two platinum electrodes. Current from a 6 volt battery is sent through the water. Hydrogen gas will be at cathode and the oxygen will be at anode.

The gases during the electrolysis can be tested with a glowing splint (which will flame up in the presence of oxygen) and burning splint (which will ignite hydrogen, causing an audible pop. H_2 gas is produced at twice the rate of O_2 gas in this reaction. Record your observations based on what happens with glowing splint at both electrodes.

17. Read the information given in the table and answer the following questions.

<i>Solution</i>	<i>pH value</i>	<i>Reaction with Phenolphthalein solution</i>	<i>Reaction with Methyl orange solution</i>
HCl	1	No colour change.	Turns into red colour.
Distilled water	7	No colour change.	No colour change.
NaOH	13	Turns into pink colour.	Turns into yellow colour.
Lemon juice	2.5	No colour change.	Turns into red colour.
NaCl	7	No colour change.	No colour change.
Baking Soda	8	Turns into pink colour.	Turns into yellow colour.

(a) List out the acids in the above table.

Ans:- HCL and Lemon Juice.

(b) What is the nature of the solution which gives pink color with Phenolphthalein solution?

Ans:-Phenolphthalein is an Acid - base indicator. It gives pink color in base solutions but no change in acid solutions.

(c) List out the neutral solutions in the above table.

Ans:- Distilled water and NaCl.

(d) Name the strongest acid and the strongest base among the given solutions.

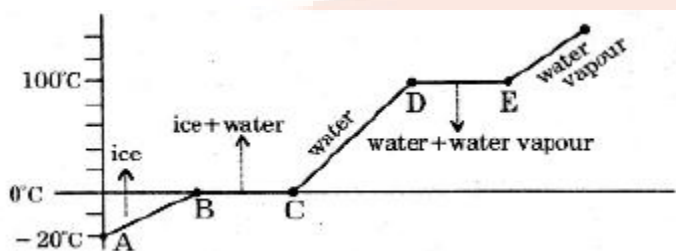
Ans:- Strongest Acid:- HCl

Strongest Base:- NaOH

OR

The graph shows the values of temperature, when ice is heated till it becomes water vapour. Observe the graph and answer the following questions.

(Note that the figure is not completely quantitative and also not to the scale. It is purely qualitative)



(a) At what temperature, ice converts into water?

Ans:- At 0°C

(b) What does $\overline{DE}$ represent?

Ans:- DE represent the water about to convert in vapour at constant temperature 100°C .

(c) What is the range of temperature of liquid water?

Ans:- 0°C to 100°C .

(d) Which part of the graph represents change of state from ice to water?

Ans:- Part AB.

Part-B

Instructions: Write the answers to the questions in this Part-B on the Question paper itself and attach it to the answer book of Part-A.

Marks : 5

Instructions:

- (i) Answer all the questions,
- (ii) Each question carries $\frac{1}{2}$ marks.
- (iii) Answers are to be written in question paper only.
- (iv) Marks will not be awarded in any case of over-writing, rewriting or erased answers.

I. Write the CAPITALLETTER (A, B,C,D) showing the correct answer for the following questions in the brackets provided against them.

1. Formation of dew is an example for

- (A) Boiling.
- (B) Melting.
- (C) Condensation.
- (D) Evaporation.

Ans.c

Dew drops are formed due to condensation of water vapors

2. $\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$ is an example for

- (A) Chemical combination.
- (B) Chemical decomposition.
- (C) Chemical displacement.
- (D) Chemical double displacement.

Ans.c

Sol. The above reaction is an example of displacement reaction as in the above reaction the elemental Zinc is displacing hydrogen from the compound of hydrochloric acid which is released in the form of hydrogen gas. The Zinc then combines with Chlorine to form Zinc Chloride.

3. Which of the following does not lie in the plane of reflection?

- (A) Reflecting surface.
- (B) Incident ray.
- (C) Normal at the point of incidence.
- (D) Reflected ray.

Ans.a

Sol. Actually the plane of reflection is the reflecting surface so it can not lie in the same plane.

4. Antacids are used

- (A) to produce acid in the stomach.
- (B) to produce water in the stomach.
- (C) to neutralize the excess base in the stomach.

(D) to neutralize the excess acid in the stomach.

Ans.d

Sol. Antacids neutralize (reduce) excess stomach acid to relieve heartburn, sour stomach, acid indigestion, and stomach upset. They can also be used to relieve the pain of stomach and duodenal ulcers. Some antacids contain simethicone, which reduces gas.

5.Principal Quantum number 3 refers to

- (A) M - main shell
- (B) f- sub shell
- (C) N - main shell
- (D) d - sub-shell

Ans.a

Sol. For the principle quantum number 3, the M shell can hold 18 electrons

$$n=3, l=0,1,2$$

6.Number of vertical columns in the modern periodic table are (As per IUPAC notation).

- (A) 7
- (B) 8
- (C) 10
- (D) 18

Ans.d

Sol. There are 18 vertical columns in the modern periodic table and these are known as groups.

7. If the refracted ray from a convex lens is travelling parallel to the principal axis, then image distance is

- (A) Equal to object distance.
- (B) Infinity.
- (C) Equal to focal length of the lens.
- (D) Equal to radius of curvature of the lens.

Ans. c

Sol. Any incident ray traveling parallel to the principal axis of a convex lens will refract through the lens and travel through the focal point on the opposite side of the lens.

8. Effect of dispersion of light among the following is

- (A) Mirage.
- (B) Blue colour of the sky.
- (C) Rainbow.
- (D) Twinkling of stars

Ans. c

Sol. The most familiar example of dispersion is probably a rainbow, in which dispersion causes the spatial separation of a white light into components of different wavelengths (different colors). However, dispersion also has an effect in many other circumstances: for example, group velocity dispersion (GVD) causes pulses to spread in optical fibers, degrading signals over

long distances; also, a cancellation between group-velocity dispersion and nonlinear effects leads to solution waves.

9. The equivalent resistance due to series connection of 10Ω and 10Ω resistors is

- (A) 5Ω
- (B) 10Ω
- (C) 0Ω
- (D) 20Ω

Ans. d

Sol. When several resistors are joined in series, then the total resistance (R_s) is equal to:

$$R_s = R_1 + R_2$$

$$R_s = 10\Omega + 10\Omega = 20\Omega.$$

10. Which of the following electrical devices works on the principle of electro-magnetic induction?

- (A) Electric fan.
- (B) Electric bulb.
- (C) Electric cooker.
- (D) L.E.D.

Ans. a

Sol. Electromagnetic Induction and its Applications.

Electromagnetic Induction or Induction is a process in which a conductor is put in a particular position and magnetic field keeps varying or magnetic field is stationary and a conductor is moving.

